

# **Noetic Learning Mathematics Contest**

**Year 2009 – 2016**



**Sample Problems  
Grades 2 – 8**

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# Noetic Learning Math Contest Sample Problems

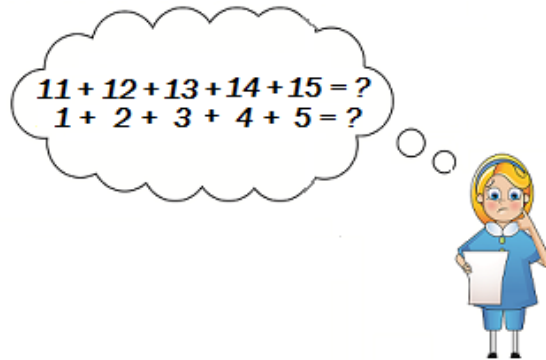
## Grade 2

- 1) Ronald had 2 quarters, 6 dimes and 3 nickels in his pocket. After he buys an ice cream cone for 75¢, how many cents does he have left?



Answer: \_\_\_\_\_ ¢

- 2) Abby is solving 2 math problems as shown in the picture. What is the difference between the answer of the first problem and the answer of the second problem?

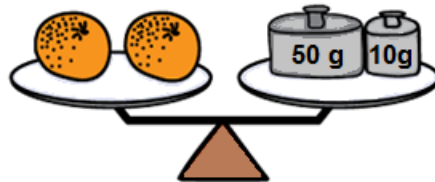


Answer: \_\_\_\_\_

- 3) Each day since Julie arrived at Granny's Orchard, she picked 10 apples, ate 3 of them and saved the rest. How many apples had Julie saved by the end of 4<sup>th</sup> day?

Answer: \_\_\_\_\_ apples

4) The following balance scale is balanced. The two oranges weigh the same. How much does each orange weigh?



Answer: \_\_\_\_\_ g

5) Today is Cancer Awareness Day. The three Petersen sisters are each wearing one piece of pink clothing. One is wearing a pink hat, one is wearing a pink coat, and one is wearing a pink scarf. Their names are Ashley, Barbara and Jamie. Ashley never wears a scarf. Barbara never wears a hat or a scarf. What pink item is Ashley wearing today?



Answer: \_\_\_\_\_

# Noetic Learning Math Contest Sample Problems

## Grade 2 Answer Keys

- 1) 50 ¢
- 2) 50
- 3) 28 apples
- 4) 30 g
- 5) hat

# Noetic Learning Math Contest Sample Problems

## Grade 3

- 1) An eraser was placed on a correctly solved addition problem. What number did it cover up?

$$\begin{array}{r} 67 \\ + \phantom{00} \\ \hline 92 \end{array}$$

Answer: \_\_\_\_\_

- 2) Victor sends 6 text messages a day to each of his 8 friends. What is the total number of text messages he sends to his friends in 3 days?



Answer: \_\_\_\_\_ text messages

- 3) How many 3-digit numbers can you make using digits 0, 2 and 9 if you can use each digit exactly once in each number?

Answer: \_\_\_\_\_ numbers

4) Jeff left home in March 1999 to tour the world. He returned exactly 40 months later. Which month of the year did he return home?

Answer: \_\_\_\_\_

5) The following scale is balanced. If the tennis ball weighs 60g and the two books weigh the same, how much does each book weigh?



Answer: \_\_\_\_\_ g

# Noetic Learning Math Contest Sample Problems

## Grade 3 Answer Keys

- 1) 25
- 2) 144 texts
- 3) 4 numbers
- 4) July
- 5) 120 g

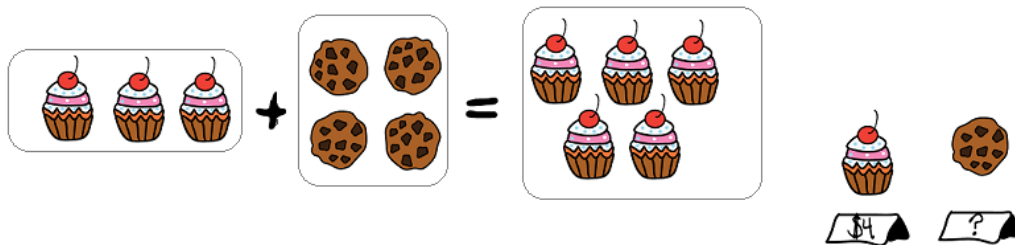
# Noetic Learning Math Contest Sample Problems

## Grade 4

1) How many 2-digit counting numbers are even?

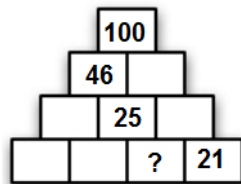
Answer: \_\_\_\_\_

2) Three cupcakes and four cookies cost the same as five cupcakes. If a cupcake costs \$4, how much does a cookie cost?



Answer: \$ \_\_\_\_\_

3) In the following number pyramid, the sum of two numbers side by side equals the number directly above them. What number should go in the box with the question mark?



Answer: \_\_\_\_\_



4) Kelsey, Rose, Gabby, Lily and Tina went to a movie together. After they sat down, Kelsey and Rose switched seats. Then Tina and Kelsey switched seats. Finally, Gabby and Tina switched seats. At the end Kelsey sat in seat A, Rose sat in seat B, Gabby sat in seat C, Lily sat in seat D and Tina sat in seat E. Who sat in seat C at the beginning?

Answer: \_\_\_\_\_

5) Jane asked about her grandma's age. Her grandma answered, "Well, I am six years younger than your grandpa. Your dad's age is half of your grandpa's age. And we both know that your age is one-quarter of your dad's age." Jane is nine years old. How old is Jane's grandma?



Answer: \_\_\_\_\_ years old

# Noetic Learning Math Contest Sample Problems

## Grade 4 Answer Keys

- 1) 45
- 2) \$2
- 3) 8
- 4) Rose
- 5) 66 years old

# Noetic Learning Math Contest Sample Problems

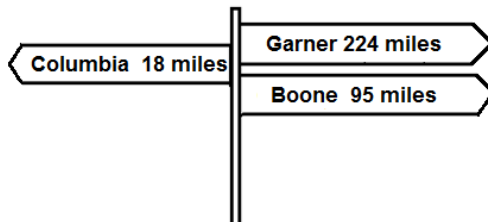
## Grade 5

- 1) From the following 9 numbers, how many different ways can you pick 3 different numbers so that their sum is 22?



Answer: \_\_\_\_\_ different ways

- 2) Carrie lives in Columbia. Trevor lives in Garner. Carrie and Trevor agreed to drive and meet in Boone today. The following sign is on the road that they will both travel on. How many more miles does Trevor have to drive to Boone than Carrie?



Answer: \_\_\_\_\_ miles

- 3) Amy and Vicki went to an orchard to pick apples. Vicki picked two more than half as many as apples as Amy picked. Together they picked 98 apples. How many apples did Amy pick?

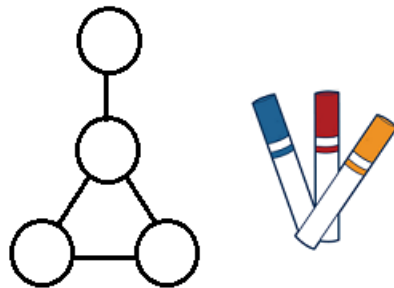
Answer: \_\_\_\_\_ apples

4) Different letters represent different digits. If "A" represents 2, what digit does "E" represent?

$$\begin{array}{r} \text{E M M} \\ + \text{E M} \\ \hline \text{M A A} \end{array}$$

Answer: \_\_\_\_\_

5) You are given 3 colors of markers: red, blue, and yellow. You are to color the circles in the following picture. You can use only one color in each circle and no two adjacent circles (connected by a line) can have the same color. How many different ways can you color the circles?



Answer: \_\_\_\_\_ different ways

# **Noetic Learning Math Contest Sample Problems**

## **Grade 5 Answer Keys**

- 1)** 2 different ways
- 2)** 16 miles
- 3)** 64 apples
- 4)** 5
- 5)** 12 different ways

# Noetic Learning Math Contest Sample Problems

## Grade 6

1) For any two whole numbers  $x$  and  $y$ , let  $x \Delta y$  be defined by  $x \Delta y = 3x - y^2$ . Therefore,  $1 \Delta 1 = 2$  and  $3 \Delta 2 = 5$ . What is the value of  $(5 \Delta 1) \Delta 6$ ?

Answer: \_\_\_\_\_

2) There are 5 candles in a row. How many different ways can you choose 2 candles to light so that the two lighted candles are **not** next to each other?



Answer: \_\_\_\_\_ ways

3) The product of the 3-digit number **NUM** and the 1-digit number **S** is 3663. If different letters represent different digits, what is the value of **NUM + S**?

$$\begin{array}{r} \text{NUM} \\ \times \quad \text{S} \\ \hline 3663 \end{array}$$

Answer: \_\_\_\_\_

**4)** Sixty-eight college basketball teams battled for the NCAA tournament championship title in the year 2014. The losing team of each game was eliminated from the tournament. What was the total number of games played in the 2014 NCAA tournament?

Answer: \_\_\_\_\_ games

**5)** Three sisters are sharing a jar of cookies. Mary took half of the cookies plus 1 more. Nancy took half of what was left plus 1 more. Pam got the remaining 3 cookies. How many cookies were in the jar originally?



Answer: \_\_\_\_\_ cookies

# Noetic Learning Math Contest Sample Problems

## Grade 6 Answer Keys

- 1) 6
- 2) 6 ways
- 3) 416
- 4) 67 games
- 5) 18 cookies



# Noetic Learning Math Contest Sample Problems

## Grade 7

- 1) Thirty-one people participated in a company picnic. People sat in groups of either 4 or 5. How many groups were there?

Answer: \_\_\_\_\_ groups

- 2) We know the following facts:

- 2 apples and 1 pear weigh more than 2 bananas and 1 pear.
- 2 pears and 1 banana weigh more than 2 apples and 1 pear.
- All apples weigh the same, all bananas weigh the same, and all pears weigh the same.



Which of the following statements is correct?

- A. One apple's weight < One banana's weight < One pear's weight
- B. One banana's weight < One apple's weight < One pear's weight
- C. One banana's weight < One pear's weight < One apple's weight
- D. One pear's weight < One banana's weight < One apple's weight

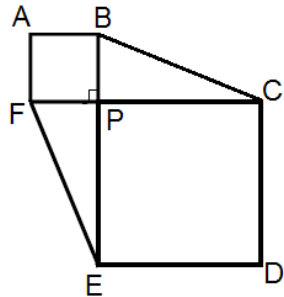
Choose either A, B, C or D as your answer.

Answer: \_\_\_\_\_

- 3) Mary and Nancy had a total of 42 crayons. After Mary gave 30% of her crayons to Nancy, Nancy had twice as many as Mary. How many crayons did Mary have originally?

Answer: \_\_\_\_\_ crayons

- 4) In the following figure, the area of square ABPF is  $25 \text{ in}^2$  and the area of square PCDE is  $144 \text{ in}^2$ . What is the area of hexagon ABCDEF?



Answer: \_\_\_\_\_  $\text{in}^2$

- 5) Peter received 10 points on a 10-question quiz. If he received 5 points for each correct answer and lost 3 points for each wrong or missing answer, how many questions did Peter answer correctly?

Answer: \_\_\_\_\_ questions

# Noetic Learning Math Contest Sample Problems

## Grade 7 Answer Keys

- 1) 7 groups
- 2) B
- 3) 20 crayons
- 4)  $229 \text{ in}^2$
- 5) 5 questions

# Noetic Learning Math Contest Sample Problems

## Grade 8

- 1) Rohit attended a 5-day math camp. Each day he solved six more math problems than he solved the previous day. Rohit solved a total of 215 math problems during the 5-day camp. How many math problems did he solve on the 1<sup>st</sup> day of the camp?

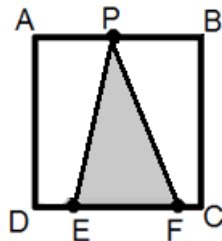
Answer: \_\_\_\_\_ math problems

- 2) Cupcakes are available in either large boxes or small boxes. There are 9 cupcakes in each large box. There are 4 cupcakes in each small box. Sam bought a total of 30 cupcakes. If a large box costs \$15 and a small box costs \$9, how much did he spend altogether?



Answer: \$\_\_\_\_\_

- 3) Square ABCD has a side of 10 inches. Points E and F are on line segment CD. Point P is on line segment AB. The area of triangle PEF is 30% of the area of square ABCD. What is the length of segment EF? (Note: The following figure is not drawn in scale.)



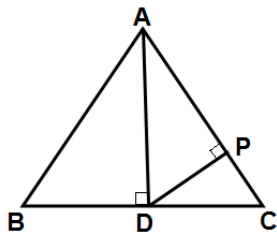
Answer: \_\_\_\_\_ inches

- 4) What is the measurement of the acute angle between the hour hand and the minute hand at 2:20 P.M.?



Answer: \_\_\_\_\_ °

- 5) The area of the following equilateral triangle ABC is  $8 \text{ in}^2$ . What is the area of triangle ADP?



Answer: \_\_\_\_\_  $\text{in}^2$

# Noetic Learning Math Contest Sample Problems

## Grade 8 Answer Keys

- 1) 31 math problems
- 2) \$57
- 3) 6 inches
- 4)  $50^\circ$
- 5)  $3 \text{ in}^2$